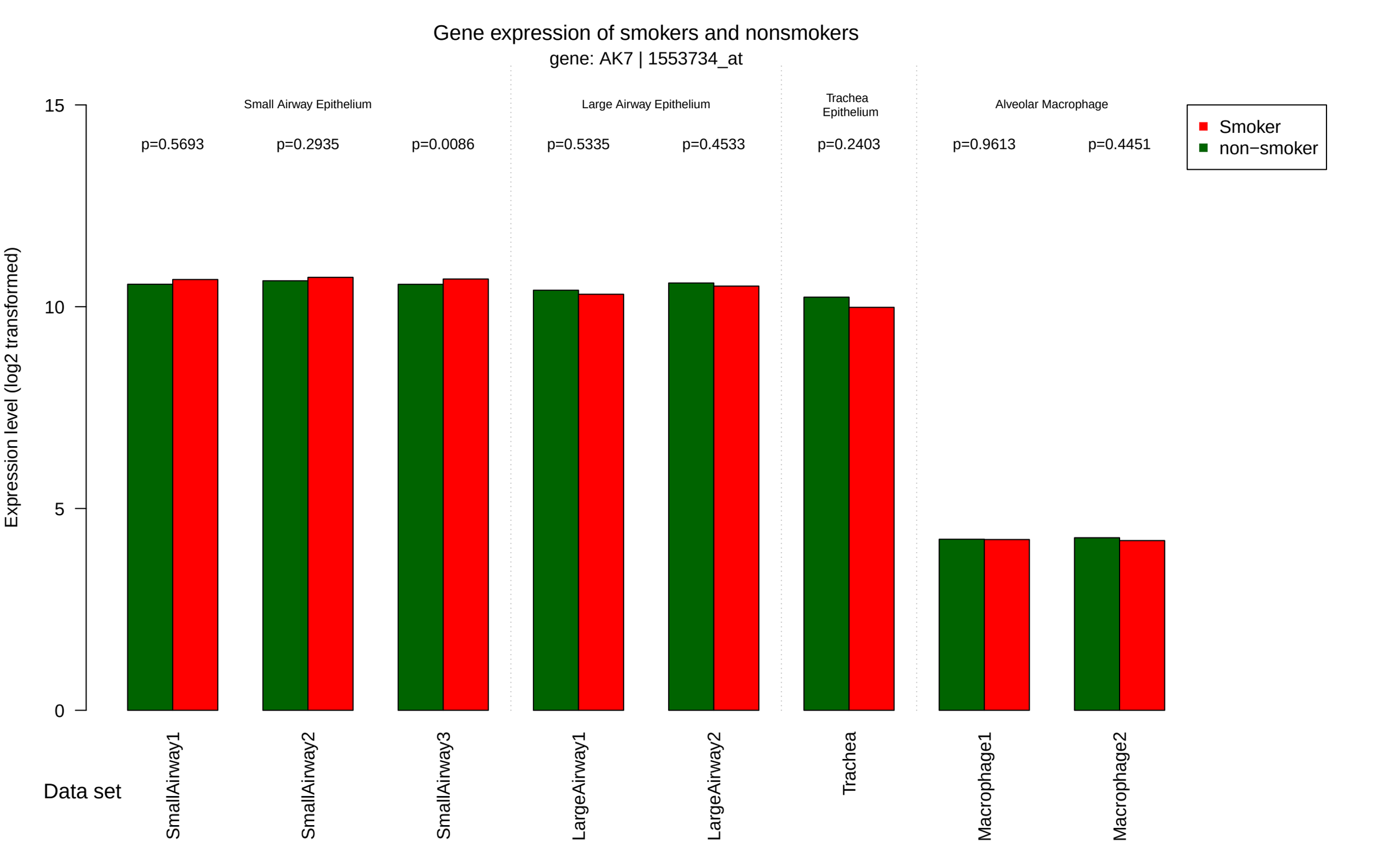
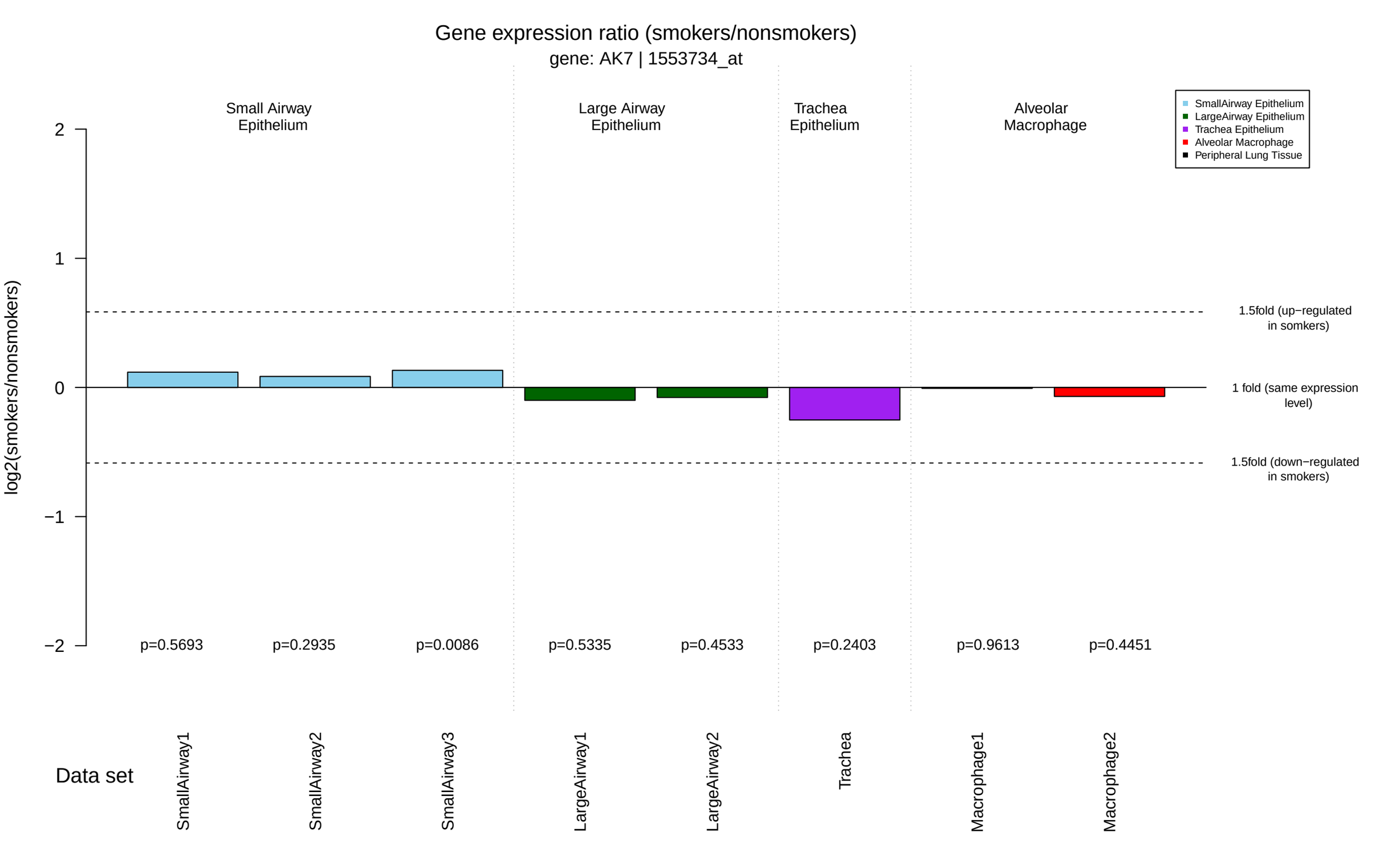
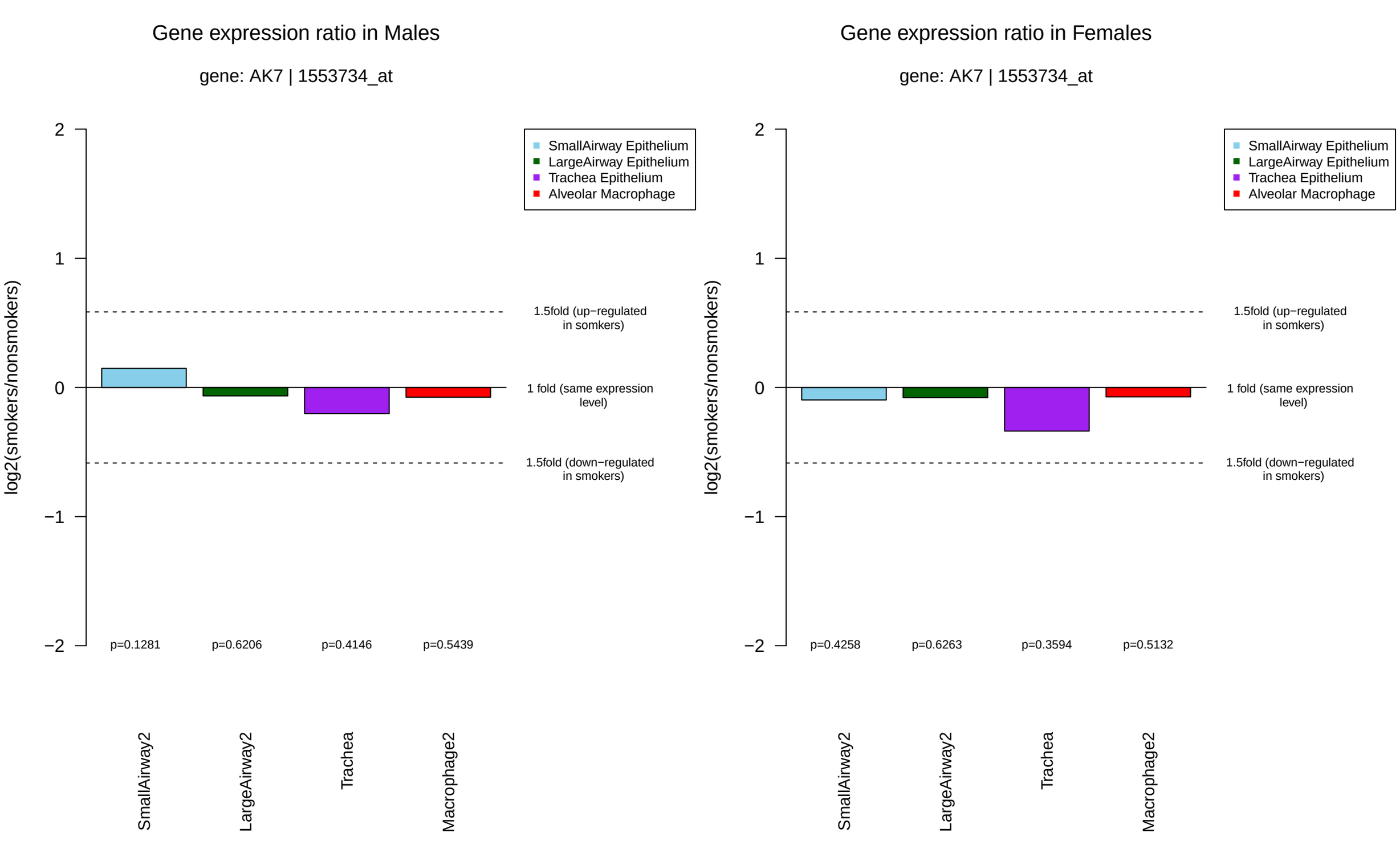
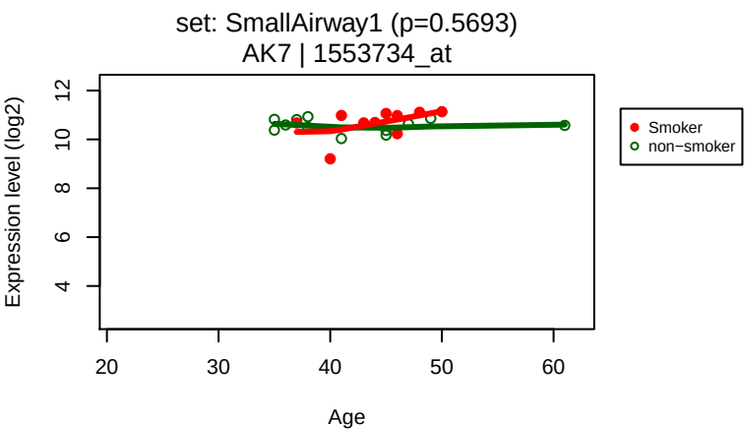
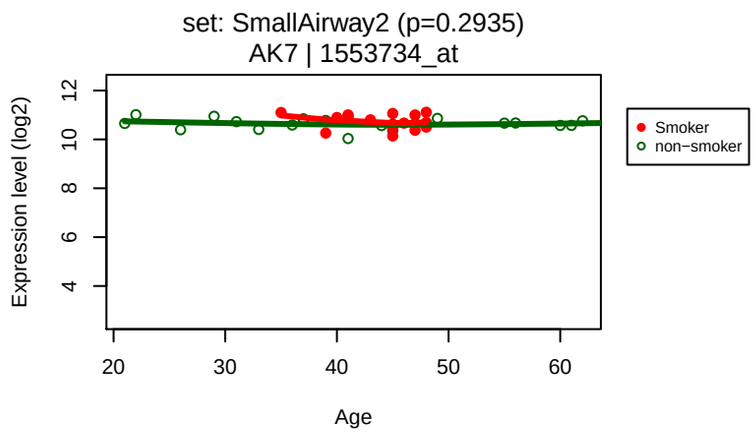
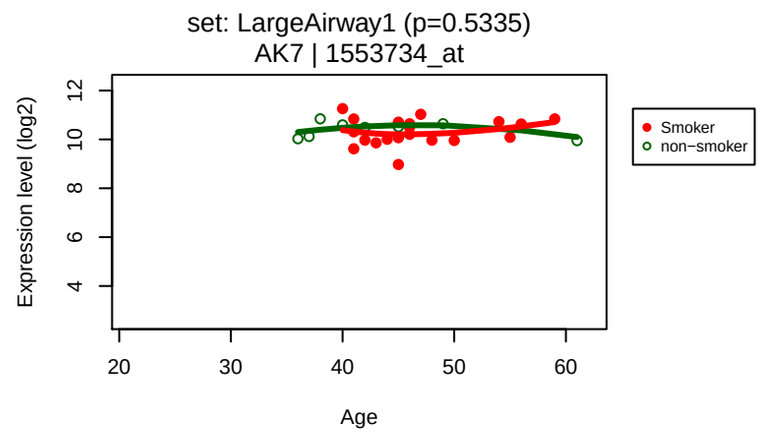

Gene expression of smokers and nonsmokers
gene: AK7 | 1553734_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5693
p=0.2935
p=0.0086
p=0.5335
p=0.4533
p=0.2403
p=0.9613
p=0.4451
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

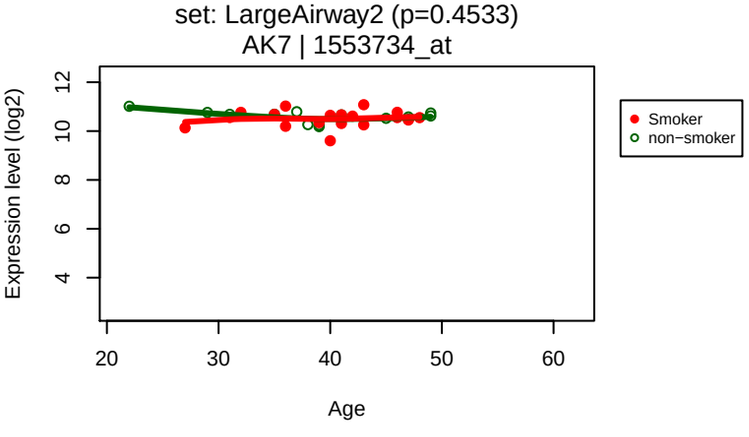

Gene expression ratio (smokers/nonsmokers)
gene: AK7 | 1553734_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5693
p=0.2935
p=0.0086
p=0.5335
p=0.4533
p=0.2403
p=0.9613
p=0.4451
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

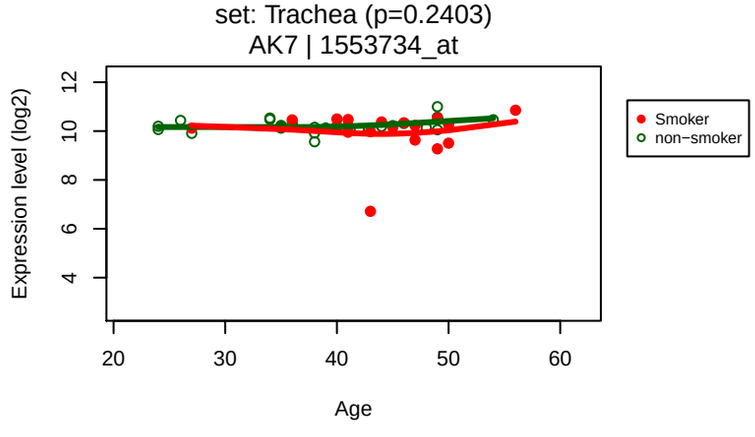

Gene expression ratio in Males
gene: AK7 | 1553734_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1281
p=0.6206
p=0.4146
p=0.5439
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AK7 | 1553734_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4258
p=0.6263
p=0.3594
p=0.5132
SmallAirway2
LargeAirway2
Trachea
Macrophage2

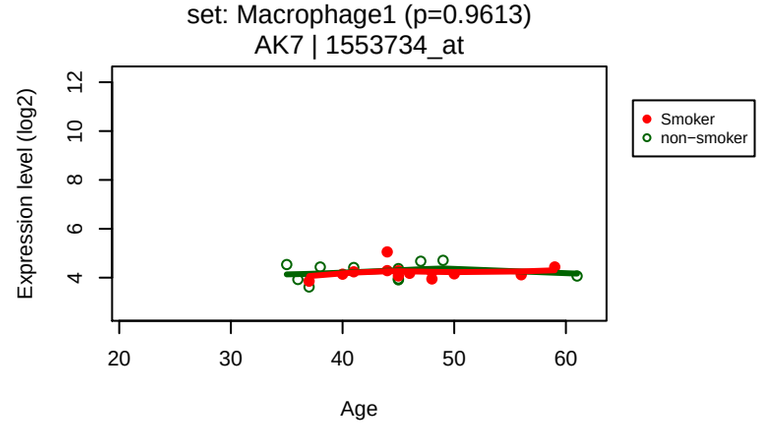

set: SmallAirway1 (p=0.5693)
AK7 | 1553734_at
Expression level (log2)
Age
Smoker
non−smoker

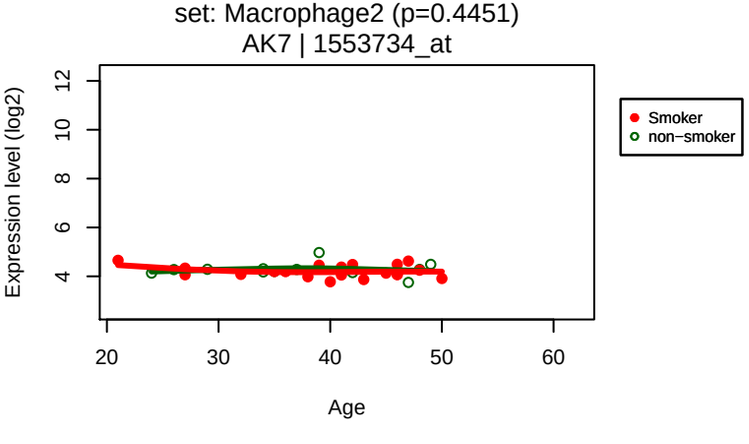

set: SmallAirway2 (p=0.2935)
AK7 | 1553734_at
Expression level (log2)
Age
Smoker
non−smoker

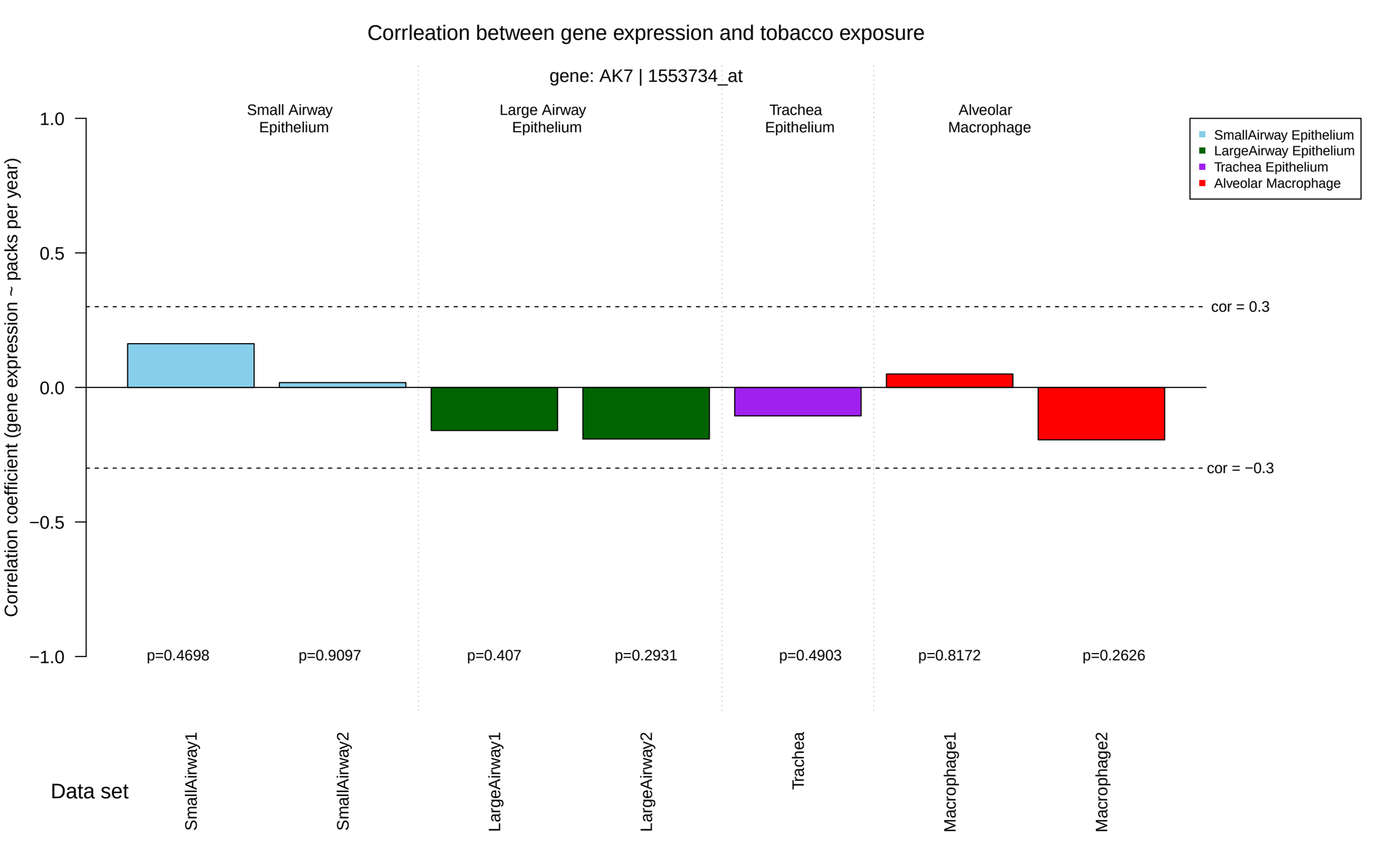

set: LargeAirway1 (p=0.5335)
AK7 | 1553734_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4533)
AK7 | 1553734_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2403)
AK7 | 1553734_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9613)
AK7 | 1553734_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4451)
AK7 | 1553734_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AK7 | 1553734_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4698
p=0.9097
p=0.407
p=0.2931
p=0.4903
p=0.8172
p=0.2626
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2